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# Antibiotic-resistant *Escherichia coli* from goat farms and the potential treatment by *Acalypha indica* L. extract

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ABSTRACT

Antibiotics have been used in both human and animal therapeutics. The present study aimed to investigate antibiotic resistance in *E. coli* isolated from goats. The antimicrobial susceptibility testing (AST) in these isolates was performed using broth microdilution. From 86 *E. coli* isolates, 2, 1, and 6 isolates were resistant to imipenem, ciprofloxacin, and colistin, respectively. Only one isolate could analyze the whole-genome of antibiotic-resistant *E. coli.* It was strain Y5.4.5 and by the MLST revealed that it belonged to ST345. This isolate harbored several antibiotic-resistant genes, including *aadA1, aadA2, blaTEM-1B, cmlA1, dfrA12, floR, qnrS1, sul1, sul2, sul3, tet*(A), conferring resistance to streptomycin, ampicillin, chloramphenicol, trimethoprim, ciprofloxacin I/R, sulfisoxazole, tetracycline. The Inc and Col-like plasmids were also identified in the isolate. There kinds of herbal, *Cinnamomum porrectum* (Roxb.) Kosterm*, Caesalpinia bonduc* and *Acalypha indica* L. were selected for extraction and analyzed the activity against antibiotic-resistant *E. coli.* The ethanol crude extract of *A. indica* L. could inhibit *E. coli* growth with the MIC of 25 mg/ml. The time-kill assay revealed that the crude extract started its inhibition activity at 4 h. It was less bactericidal but bacteriostatic was found. Our study presents the whole genome of antibiotic-resistant *E. coli* strain Y5.4.5 isolated from goats, and the *A. indica* L. extract might be combined with other antibiotics or herbal extracts as alternative medicine.

## 1. Introduction

Antimicrobial resistance (AMR) is global public health worrisome due to limitations in therapeutic options. Among AMR, multidrug resistance *Enterobacteriaceae,* especially *Escherichia coli* can be isolated from humans, animals, and the environment, leading to public health problems for both humans and animals. In addition, the emergence of multidrug resistance *E. coli* in animals has adverse health impacts on human health due to physical proximity and frequent contact with the owners (Madec et al., 2017). Therefore, understanding the occurrence and characteristics of antibiotic-resistant *E. coli* is essential for both human and veterinary health perspectives.

*E. coli* are commensal bacteria in the gut of both humans and animals. However, six pathogenic phenotypes can cause the disease in humans such as Enteropathogenic *E. coli* (EPEC)*, Enterohaemorrhagic E. coli* (EHEC)*, Enterotoxigenic E. coli* (ETEC)*, Enteroaggregative E. coli* (EAEC)*, Enteroinvasive E. coli* (EIEC)*, Diffusely adherent E. coli* (DEEC) (Kaper et al., 2004). The common diseases in humans are enteritis, community- and hospital-acquired urinary tract infection (UTI), septicemia, postsurgical peritonitis, and neonatal meningitis, while in animal, diarrhea are commonly found (Ramos et al., 2020). Therefore, multidrug-resistant (MDR) *E. coli* in animals is important because it can be transmitted to humans.

*E. coli* has been implemented in various national antibiotic resistance surveillance programs reporting data with human clinical, livestock carriage, milk and meat product, and clinical veterinary (Anjum et al., 2021). In Thailand, the report of AMR *E. coli* in gastrointestinal tracts of the adults cultured was 66.5% (Khamsarn et al., 2016), and AMR *E. coli* from tertiary hospitals showed a high resistance rate to ciprofloxacin (85.26%) (Bubpamala et al., 2018). In Northern Thailand, AMR *E. coli* was isolated from poultry farmers (50%), backyard chickens (24.9%), backyard ducks (36.6%), and the environment (25.0%) (Tansawai et al.,

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2019). MDR *E. coli* was present in broiler farms (60.4%) and pig farms (36.7%) in Chiang Mai-Lamphun province (Nuangmek et al., 2018; Rodroo et al., 2021). In addition, the study in dairy goat farms showed 78.3% of MDR *E. coli* with primary resistance to streptomycin (65.6%) and tetracycline (30.0%) (Prapasawat and Intarapuk, 2021). Antibiotic-resistant mechanisms in *E. coli* are mediated by alteration or loss of porin, hyperproduction of efflux pump, target modification, and production of specific enzymes. Plasmid-mediated resistance genes, including *rmtB* (conferring aminoglycoside resistance), $bla_{CTX\text{-}M}$ (conferring beta-lactam resistance) and *mcr-1* (conferring colistin resistance) were determined in MDR *E. coli* isolated from goat farms in Saudi Arabia (Shabana and Al-Enazi, 2020) and pig farms in Chiana (Liu et al., 2016b).

Treatment of MDR *E. coli* infection relies on certain antibiotics, including colistin, fosfomycin, tigecycline, and aminoglycosides (Perez et al., 2016). However, significant concerns have been regarding the limited efficacy of these options because of their pharmacologic characteristics and adverse events. Therefore, several herbal substances have been studied as alternative treatment methods for many MDR bacteria. The stream distilled oil extract from *Cinnamomum porrectum* (Roxb.) Kosterm showed strongest activity against *Streptococcus mutans* (MIC 0.01 mg/ml) followed by *Candida albicans* and dermatophyte (0.5–1.0 mg/ml), *Bacillus subtilis* (2 mg/ml), and susceptible strains of *S. aureus* (4–16 mg/ml) (Phongpaichit et al., 2007). Crude methanol extract of *Caesalpinia bonduc* Roxb leaves has significant antibacterial, against Gram positive bacteria such as *Bacillus subtilis, B. cereus, B. megaterium* and *S. aureus* as well as Gram negative bacteria including *Salmonella typhi, Shigella dysenteriae, Pseudomonas aeruginosa, Klebsiella spp.* and *E. coli* (Billah et al., 2013). *A. indica* L. showed antibacterial activity against Gram-positive organisms (*S. aureus*, *S. epidermidis*, *B. cereus, Streptococcus faecalis*) and Gram-negative (*Klebsiella pneumoniae*, *E. coli*, *Proteus vulgaris*, *P. aeruginosa*) organism with minimum inhibitory concentrations (MIC) between 0.156 and 2.5 mg/ml. (Govindarajan et al., 2008).

Goat production in Thailand has been growing intensively as halal food production. However, the situation of MDR *E. coli* in goat meat production in Thailand has not been reported. This research aims to study risk factors, molecular characteristics and the prevalence of antibiotic-resistant *E. coli* in goat production. In addition, the efficiency of the herbal extracts against the growth of antibiotic-resistant *E. coli* was also investigated. Elucidation of risk factors and characteristics of AMR *E. coli* would reduce the dissemination of the pathogen to consumers and assure food safety of the goat meat product.

## 2. Materials and methods

### 2.1. Sampling and data collection

A total of 500 specimens were derived from 50 goat farms in 5 provinces in Southern Thailand during January - July 2021. The rectal swabs of goats were collected and kept in Stuart's transport media (Oxoid, Basingstoke, Hampshire, UK) during transportation to the laboratory. The history of antibiotic use on each farm was recorded by interview.

### 2.2. *E. coli* isolation and identification

All rectal swab samples were pre-enrichment by used buffered peptone water (BPW) broth (Oxoid Ltd., Basingstoke, Hampshire, UK) for at 37 °C for 12 h. Then cultured on Eosin Methylene Blue Agar (EMB) (Himedia, India) for selecting suspected (metallic sheen) colony. Biochemical test, IMViC were identified *E. coli* as following; indole and Methyl Red (MR) positive, lactose fermentation, and Voges Proskauer (VP) and citrate negative.

Confirmation by cultured suspected colony on MacConkey agar (Sigma-Aldich® Darmstadt, Germany) plates followed by incubation at 37 °C for 24 h. Five isolated pink colonies from each plate were collected and transferred to Tryptic soy agar (Sigma-Aldich® Darmstadt, Germany), supplemented with 6 μg/l carbapenems (Sigma-Aldrich, St Louise, MO, USA) which the purposed to screen beta-lactam antibiotic resistant characteristics in bacterial isolates (Chukamnerd et al., 2021; Kaewnirat et al., 2022) and then incubated at 37 °C overnight. *E. coli* isolates were confirmed by using the presence of the *uidA* gene that coded beta-glucuronidase activity (Heininger et al., 1999). Bacterial genomic DNA was extracted by using the colony boiling method. Briefly, a full loop of bacteria was transferred to 100 μl of deionized water and boiled at 100 °C for 10 min, and placed on ice for 15 min. PCR mixture contained 1 ×PCR buffer (0.25 mM dNTP, 1.5 mM $MgCl_2$), 1 μM of each forward and reverse primer, 1 U Taq polymerase (bioline, UK), and 5 μl of DNA template in a total volume of 25 μl. The reaction was carried out in Bio-Rad T100 Thermal cycler (Biorad, California, USA) as follows: 95 °C for 3 min; 35 cycles of 1 min at 94 °C, 50 s at 60 °C, and 1 min at 72 °C; 10 min at 72 °C. The PCR products were investigated by agarose gel electrophoresis and visualized by UV.

### 2.3. Antimicrobial susceptibility testing

The minimum inhibitory concentrations (MICs) of imipenem, ciprofloxacin, meropenem, and colistin (Sigma-Aldrich, St Louise, MO, USA) were determined using the broth microdilution method and interpreted with Clinical and Laboratory Standards Institute (CLSI) interpretation guidelines, 2018 (Clinical and Laboratory Standards Institute CLSI, 2019). Performance standards for antimicrobial susceptibility testing: twenty-seventh informational supplement M100-S28. Wayne). *E. coli* ATCC 25922 was used as a control.

### 2.4. Herbal extraction and analysis of antioxidant properties

Fresh herbal leaves, *Cinnamomum porrectum (Roxb.) Kosterm, Caesalpinia bonduc (L.) Roxb.*and *A. indica* L. were cleaned and then dried at 60–65 °C for 15 min. Dried herbal leaves were mashed with a blender. One thousand one hundred thirty-five grams of herbal powder were mixed with 2300 milliliter of 95% ethanol in the glass jar tightly closed for 3 days and stirred the mixture daily. All herbal extracts were sent to analyze antioxidant properties according to the protocol of 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging activity and 2,2′-azino-bis (3-ethylbenzthiazoline-6-sulfonic acid) (ABTS) radical scavenging activity (Musarium Co. LTD., Songkhla, Thailand).

### 2.5. Antibacterial assay

The herbal extracts were evaluated for their antibacterial activities against antibiotic-resistant *E. coli* by using the agar dilution method. Initially, the compounds were dissolved in 1:3 of 95% ethanol and distilled water to prepare the stock solutions. Then the tested compounds were prepared in Mueller-Hinton agar (MHA) (Himedia, India) to obtain the required concentrations (25–1.56 mg/ml). The bacterial suspension was adjusted with sterile saline to a concentration of $1 \times 10^5$ CFU/ml and 4 dropped (2 μl) into MHA supplemented with each compound, followed by incubation at 37 °C for 18–24 h. The experiment was done in triplicate.

### 2.6. Time-kill assay

Time-kill assay for antimicrobial agents was performed using CLSI guidelines. Briefly, a Time-kill assay of a sensitive strain was performed. The herbal compound was prepared in Mueller-Hinton broth (Himedia, India) containing 25, 35, and 50 mg/ml. The *E. coli* suspension was added to obtain a final concentration ($1 \times 10^5$ CFU/ml). The bacterial number was measured by using MacConkey agar at 2, 4, 6, and 24 h. This experiment was performed in triplicate.

*2.7. DNA extraction and whole-genome sequencing analysis*

Nine resistance (6R Colistin, 2R Imipenem and 1R Ciprofloxacin) and 5 intermediate resistances (4I colistion, 1I Ciprofloxacin) isolated were selected for total genomic DNA extraction by using PrestoTM Mini gDNA Bacteria Kit (Geneaid, Taiwan) according to the manufacturer's instructions and observed through agarose gel electrophoresis. DNA concentrations were determined by the NanoDrop 8000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA). Whole-genome sequencing was performed using the BGISEQ-500 platform (Beijing Genomics Institute, China), which produced 150-bp paired-end reads. All sequence reads were *de novo* assembled using SPAdes (Genome Assembler v3.13.0) (Bankevich et al., 2012). Genomes were annotated using the Prokka (Prokaryotic Genome Annotation Pipeline version 1.11 (Seemann, 2014).

Genes conferring antibiotic resistance were predicted using ResFinder-4.1 in the Center for Genomic Epidemiology (CGE) (https://cge.cbs.dtu.dk/services/ResFinder/); 90% identity threshold (I.D.) and 80% minimum length (Zankari et al., 2012), and Comprehensive Antibiotic Resistance Database (CARD) version 5 (https://card.mcmaster.ca/) (Alcock et al., 2020). The replicon types of plasmids were identified using PlasmidFinder-2.1 in CGE, (https://cge.cbs.dtu.dk/services/PlasmidFinder/); 90% identity threshold (I.D.) and 80% minimum length (Carattoli et al., 2014).

Multilocus sequence typing (MLST) was determined using the Pasteur institute MLST scheme (Dauga, 2002) from WGS data of all isolates.

## 3. Results

A total of 500 fecal samples were collected from goats on 49 farms in 5 provinces in Sothern of Thailand, including Songkhla, Satun, Yala, Pattani, and Narathiwat (Fig. 1). Forty-two farms (85.71%) have used antibiotics, and the remainder, seven farms (14.29%) have not used antibiotics in goats. The frequently used antibiotics were beta-lactams (penicillin, penicillin-hydrostreptomycin), sulfonamide (sulfa-trimethoprim), tetracycline (oxytetracycline), and fluoroquinolone (enrofloxacin). Antibiotics were used to treat diarrhea, fever, and mastitis. In addition, some farms used antibiotics for metritis prevention by injecting the female goat after delivery (Table 1).

Of 500 fecal samples, 2251 lactose-positive bacterial isolates were collected (389, 392, 480, 490, and 500 from Satun, Songkhla, Pattani, Yala, and Narathiwat, respectively). One hundred and nine of them (109/2251; 4.84%) were able to grow on tryptic soy agar, supplemented with 6 μg/l meropenem. These isolates were suspected of ampicillin-resistant Enterobacteriaceae. By PCR targeting a specific uidA gene, 86 out of 109 isolates were confirmed as *E. coli*. These isolates were obtained from Songkhla (72), Satun (7), and Yala province (7) (Table 2). Susceptibilities to imipenem, ciprofloxacin, meropenem and colistin were determined based on MIC values (Table 3). The results showed that all isolates were susceptible to meropenem. Six, one and two isolates were resistant to colistin, ciprofloxacin, and imipenem, respectively.

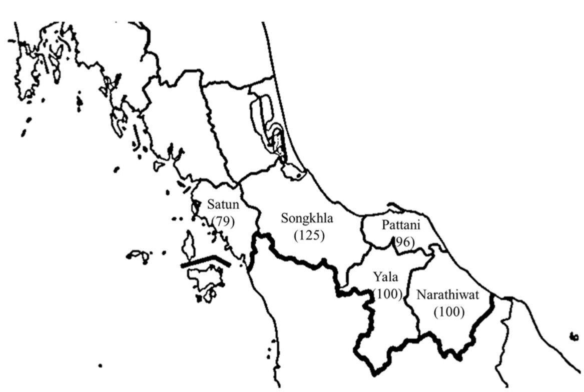


**Fig. 1.** Location of the sampled provinces and the number of samples taken in each province in Southern Thailand.

The analysis of antioxidant properties of herbal extracts indicated that *C. porrectum (Roxb.) Kosterm* possessed the highest $IC_{50}$ of DPPH and ABTS at 0.199 ± 0.002 mg/ml and 0.087 ± 0.002 mg/ml, respectively. The indicated constituent's antioxidant, phenolic, and flavonoid were 0.236 ± 0.008 mg GAE/g and 0.132 ± 0.007 g CE/g, respectively. *A. indica* L. showed $IC_{50}$ of DPPH 3.503 ± 0.013 mg/ml and $IC_{50}$ of ABTS 0.753 ± 0.025 mg/ml. The lowest antioxidant property was found in *C. bonduc (L.) Roxb*, the $IC_{50}$ DPPH and ABTS were 6.837 ± 0.070 mg/ml and 0.682 ± 0.009 mg/ml. Phenolic and flavonoid were 0.037 ± 0.001 mg GAE/g and 0.0.034 ± 0.002 g CE/g.

Inhibiting activity of the crude herbal extract against *E. coli* was found only in *A. indica* L. extract. The compound possessed a strong activity with a MIC of 25 mg/ml against only *E. coli* strain Y5.4.5 but not the others (Fig. 2). The results from the time-kill assay demonstrated that the 50 mg/ml compound inhibited the bacterial growth of more than 2 log colony-forming units (CFU/ml) at 4 h of incubation compared to untreated group (Fig. 3).

*E. coli* strain Y5.4.5 from Yala with ciprofloxacin resistance was molecularly characterized by whole genome sequencing since the *A. indica* L. extract inhibited this isolate. Multilocus typing based on housekeeping genes indicated that *E. coli* strain Y5.4.5 belonged to ST345. Its genome consisted of resistant genes, including *aadA1, aadA2, blaTEM-1B, cmlA1, dfrA12, floR, qnrS1, sul1, sul2, sul3, tet(A)*. Those genes demonstrated resistance to several antibiotics, including streptomycin, ampicillin, chloramphenicol, trimethoprim, ciprofloxacin I/R, sulfisoxazole, and tetracycline. The Y5.4.5 also harbored plasmid Col440II, IncFIA(HI1), IncHI1A, and IncHI1B(R27). The datasets used for this study can be found in the NCBI Bioproject PRJNA865859.

## 4. Discussion

This research showed a low prevalence of MDR *E. coli* even with the use of antibiotics in goat meat production in Thailand. However, the proportion of farms that use antibiotics is increasing and might be at high risk of antimicrobial resistance in the future. The finding of colistin resistance *E. coli* might be cautious to the public health because colistin was a last-resort drug for treating MDR Gram-negative bacteria infection in humans. Several bacteria, including *Pseudomonas aeruginosa, Acinetobacter baumannii, and Enterobacteriaceae members,* such as *Escherichia coli, Salmonella* spp.*, and Klebsiella* spp. have acquired colistin resistance (Aghapour et al., 2019). Mechanisms of colistin resistance consist of modification of LPS and disrupting the organization of the LPS-phospholipid bilayer, changing its permeability (Li et al., 2020). In addition, the *mcr-1* located on plasmids was present in *E. coli* isolated from pigs and meat in Southern China (Liu et al., 2016a). This gene encodes the pEtN transferase enzyme that catalyzes the addition of pEtN to lipid A for decreasing colistin affinity (Wang et al., 2020). Sequence analysis indicated that the *mcr-1* was absent in *E. coli* strain Y5.4.5, correlating to its colistin susceptible character. Multilocus typing based on seven housekeeping genes noted that *E. coli* strain Y5.4.5 belonged to ST345. This sequence type (ST 345) was firstly isolated from commercial chicken meat in Brazil (Casella et al., 2017). Multiple resistance genes conferring resistance to various classes of antimicrobial agents were present in the Y5.4.5. It harbored *aadA1, aadA2, blaTEM-1B, cmlA1, dfrA12, floR, qnrS1, sul1, sul2, sul3, tet(A),* which conferred resistance to streptomycin, ampicillin, chloramphenicol, trimethoprim, ciprofloxacin, sulfisoxazole and tetracycline. This result suggested multidrug resistance of Y5.4.5. Interestingly, our isolate did not carry the $bla_{CTX}$ gene, which was found in the ST345 from Brazil. The Y5.4.5 carried Col440II, IncFIA(HI1), IncHI1A and IncHI1B(R27) plasmid. The Inc-type plasmids and Col-like plasmids have been reported as multidrug resistance plasmid with the carriage of the *bla* genes. This plasmid has been frequently found in *E. coli* and disseminated for many decades

**Table 1**
The results of a survey on the antibiotic used in goat farms in 5 provinces in Southern of Thailand.

| Province | Number of farms | Number of farms that have used antibiotics | Number of farms that have not used antibiotics | Antibiotics | Purpose of antibiotic use |
|---|---|---|---|---|---|
| Songkhla | 12 | 8 | 4 | Penicillin, enrofloxacin, oxytetracycline, penicillin-dihydrostreptomycin | Treatment of fever, wound infection |
| Satun | 10 | 7 | 3 | Penicillin, oxytetracycline | Treatment of infection and injection after delivery |
| Pattani | 7 | 7 | 0 | Penicillin, oxytetracycline, sulphamethoxazole-trimethoprim, penicillin-dihydrostreptomycin | Treatment of diarrhea, mastitis and respiratory infection |
| Yala | 10 | 10 | 0 | Penicillin, enrofloxacin, penicillin-dihydrostreptomycin | Treatment of fever, wound infection |
| Narathiwat | 10 | 10 | 0 | Penicillin | Treatment of infection |
| Total | 49 | 42 | 7 | | |

**Table 2**
Bacterial collection and identification in each province.

| Province | Total farm | Total sample | Total *Enterobacteriaceae* isolates | antibiotic resistance isolates | The isolates of *E. coli* |
|---|---|---|---|---|---|
| Satun | 10 | 79 | 389 | 6 | 7 |
| Songkhla | 12 | 125 | 392 | 93 | 72 |
| Yala | 10 | 100 | 480 | 10 | 7 |
| Pattani | 7 | 96 | 490 | 0 | 0 |
| Narathiwat | 10 | 100 | 500 | 0 | 0 |
| Total | 49 | 500 | 2251 | 109 | 86 |

**Table 3**
Antibiotic susceptibility test of *E. coli* upon 4 antibiotics (Imipenem, Ciprofloxacin, Meropenem, and Colistin).

| Province | *E. coli* isolates | MIC results | | | |
|---|---|---|---|---|---|
| | | Imipenem | Ciprofloxacin | Meropenem | Colistin |
| Satun | 7 | – | 1(R) | – | 4(I) |
| Yala | 7 | – | 1(I) | – | – |
| Songkhla | 72 | 2(R) | – | – | 6(R) |
| Total | 86 | 2 | 2 | 0 | 10 |

R: Resistant,
I: Intermediate resistant

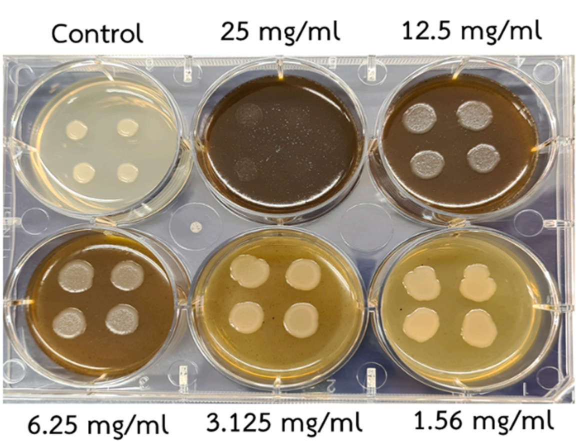


**Fig. 2.** The inhibitory ability of *A. indica* L. crude extract against *E. coli* strain Y5.4.5. Control was MHA media without herbal extracts.

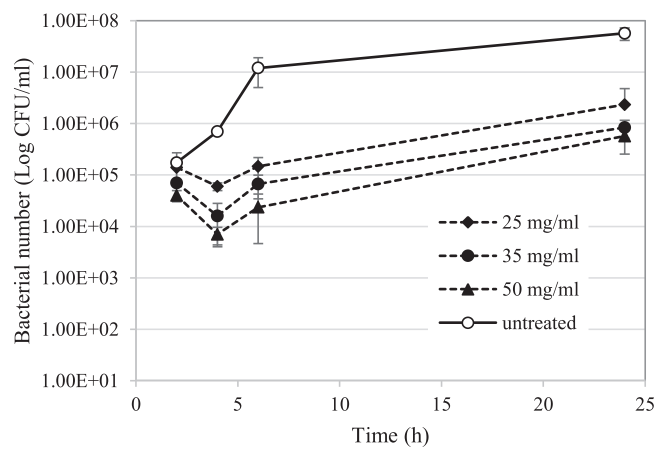


**Fig. 3.** : The time-kill assay of *A. indica* L. against *E. coli* strain Y5.4.5.

(Bonnin et al., 2012; Diaconu et al., 2020; Fiett et al., 2014; Pitart et al., 2015; Xu et al., 2019).

According to the study of antibiotic resistance in dairy goats in Thailand, the prevalence of MDR *E. coli* was 23.9%, and these isolates harbored class 1 and 2 integrons. These integrons have been associated with streptomycin, tetracycline, enrofloxacin, and sulfamethoxazole/trimethoprim resistance (Prapasawat and Intarapuk, 2021). Fifty-one percent of MDR *E. coli* were reported from healthy and diarrheal sheep and goats from Egypt. These isolates carry *rmt*B gene conferring aminoglycoside resistance and various *bla*$_{CTX-M}$ genes producing extended-spectrum beta-lactamases (ESBL) (Shabana and Al-Enazi, 2020). In the study of retail goat meat products in India, 16 of 22 isolates that carried intI1 produced carbapenemase and/or carried ESBL genes (Singh et al., 2020).

The evaluation of antioxidant properties revealed that *C. porrectum (Roxb.) Kosterm* consisted of antioxidant, DPPH and ABTS and indicated the constituent's antioxidant (Phenolic and flavonoid) higher than *C. bonduc* and *A. indica* L. Our study showed that only *A. indica* L. inhibited the growth of ciprofloxacin-resistant *E. coli* strain Y5.4.5. This compound did not inhibit the growth of other isolates. The extract of *A. indica* L. has exhibited inhibitory activity against the growth of many pathogenic bacteria. The leaves extract inhibited the growth of Gram-positive bacteria such as *Staphylococcus aureus, Staphylococcus epidermidis, Bacillus cereus, Streptococcus faecalis* and Gram-negative bacterium, *Pseudomonas aeruginosa* (Govindarajan et al., 2008). The extract has also demonstrated the broad spectrum against other pathogenic bacteria, including *Bacillus subtilis, E. coli, Salmonella typhi, Shigella flexneri, Klebsiella pneumoniae, andVibrio cholerae* (Saranraj et al., 2010). While, its flower extract inhibited the growth of *Enterohaemorrhagic Escherichia coli* (EHEC) 0157: H7 with a MIC of 200 mg/ml (Nikmah et al., 2018). The ethanolic fraction exhibited superior activity against MDR pathogens (ESBL producing *E. coli* and Carbapenem-resistant *E, coli*) isolated from blood cancer patients with a MIC of 1024–512 μg/ml (Suresh et al., 2021). Our study revealed higher concentration for inhibited *E. coli* strain Y5.4.5 with a MIC of 25 mg/ml, which might because of difference concertation ingredient due to the area of *A. indica* L. collection. Even though the extract inhibited the bacterial growth of more than 2 logs at 4 h incubation, the bacterial culture regrew and reached log 6 at the end of the experiment while untreated group reached log 8. This result implied that the compound might not have bactericidal activity. The explanation of this phenomenon needs to be clarified in further study.

As a result, the prevalence of MDR *E. coli* was very low even high number of farms that have used antibiotics. The *A. indica* L. leave extract can reduce bacterial growth to less than 2 log CFU/ml at concentrations of 25 mg/ml. Although the compound was less bactericidal but bacteriostatic was found, it may be combined with other antibiotics or herbal extracts as alternative medicine. The result of our study will implement antibiotic resistance surveillance programs in goat production and reinforce a setup of farm standards to follow good practice on the use of veterinary drugs. In addition, goat farm management and reducing the contamination of antibiotic-resistant bacteria in goat production will be encouraged to promote organic goat farms in the future. The limitation of this study was lack of goat meat samples because of there was no goat slaughter house in the study areas, it has only household killing and sell according to preorder only.

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### *Ethic approval*

This project has been approved by the Institutional Animal Care and Use Committee, Prince of Songkla University (Approval no. MHESI 68014/384).

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